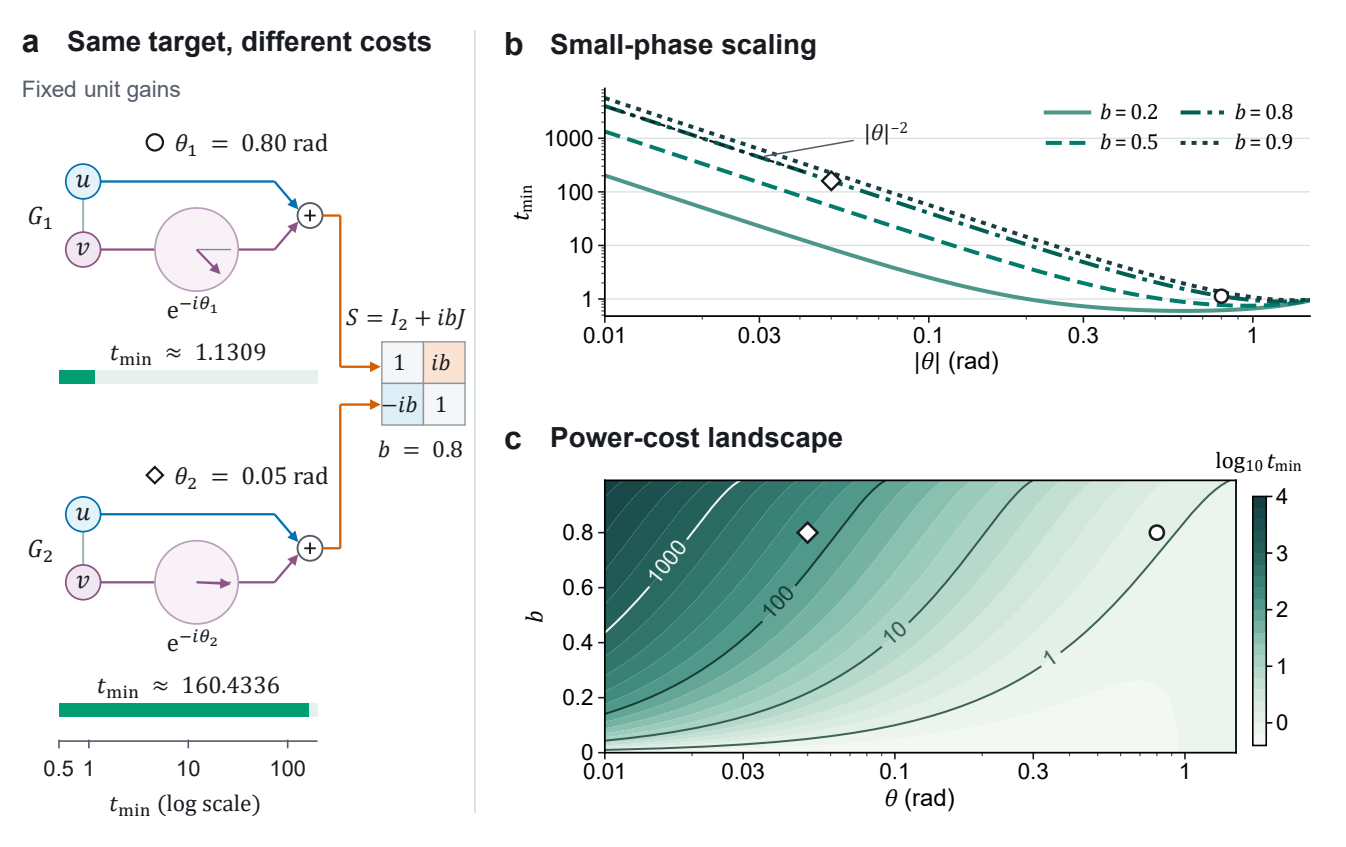

a Same target, different costs
Fixed unit gains
○ $\theta_1 = 0.80$ rad
G_1
u
v
$e^{-i\theta_1}$
$t_{min} \approx 1.1309$
$S = I_2 + ibJ$
1 ib
$-ib$ 1
$b = 0.8$
◇ $\theta_2 = 0.05$ rad
G_2
$e^{-i\theta_2}$
$t_{min} \approx 160.4336$
0.5 1 10 100
t_{min} (log scale)
b Small-phase scaling
t_{min}
$|\theta|^{-2}$
$b = 0.2$
$b = 0.5$
$b = 0.8$
$b = 0.9$
1000 100 10 1
0.01 0.03 0.1 0.3 1
$|\theta|$ (rad)
c Power-cost landscape
$\log_{10} t_{min}$
4 3 2 1 0
b
0 0.2 0.4 0.6 0.8
1000 100 10 1
0.01 0.03 0.1 0.3 1
θ (rad)